

Spatio-Temporal Agnostic Deep Learning Modeling of Forest Fire Prediction Using Weather Data

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Abstract—This research provides a spatio-temporal agnostic framework based on subsampling to generate generic deep learning models using publicly available weather data and to predict the probability of forest fire and severity. The aim is to show that this framework can be used to subsample and generate a balanced dataset for generic deep learning models to improve predictions for forest fires. The framework works for binary classification and regression deep learning models. It also works with limited variations between fire and non-fire data. Using this framework, 45 of the binary classification models built produced an $F1Score$ greater than 0.95 while 35 of 54 regression models produced an R^2Score greater than 0.91.

Index Terms—Forest Fires, Fire Weather Index, Machine Learning, Deep Learning, Data Sampling, Dataset Balancing, Big Data Analytics, Data Mining.

I. INTRODUCTION

The Annual Forests Report, 2022 [1] states that Canada ranks third in the world in forest cover with almost 362 million hectares accounting for 205,365 direct jobs and 2.3 billion dollars in revenue. In 2021, 6,596 forest fires burned 4,307,520 hectares in Canada. This was above the annual average and is a cause for concern. Therefore, it is important to detect, monitor and predict forest fires. Research on forest fire prediction [2]–[4] takes into consideration lightning caused forest fires [5]–[8]. The Canadian Forest Fire Weather Index (FWI) System is a part of the Canadian Wildland Fire Information System (CWFIS) which provides the fire danger level values for the entire country. Forest fires occur often but they are few and far apart in the same region. This leads to more non-fire points than fire points for forest fire datasets. The data available for research on forest fire has more than 100,000 non-fire points for one fire point.

The objectives of this paper are to: (1) develop a balanced dataset framework, and thereby, provide a spatio-temporal agnostic subsampling for imbalanced forest fire dataset, and (2) build generic models using the Canadian weather station data to predict the probability and severity (area burnt) of the fire. Temporal agnostic data is used to ensure that the

processed data does not pick up temporal information to make its predictions. Using a large spatially agnostic dataset in which data was spread across multiple regions helped build a model that is generic enough to be used in diverse weather conditions. Binary classification models were used to predict the probability of fire while regression models were used to predict the severity of the fire. The resulting models trained using this subsampling framework learned the fire weather behavior over the sampling period. The results show that 45 binary classification models produced an $F1Score > 0.95$. 35 regression models trained returned an $R^2Score > 0.91$.

The rest of the paper is organized as follows: Section II presents the related work in data collection, sampling, forest fire detection, and forest fire prediction. Section III discusses the datasets and methodology is presented in Section IV. The experiments along with the results are presented in Section V and Section VI gives concluding remarks.

II. LITERATURE REVIEW

Subsampling is a technique used to balance datasets when there is imbalanced data. Ai et al. [9] derived optimal subsampling probabilities for generalized linear regression models. In [10], Yao and Wang showed how optimal subsampling methods are effective in reducing the computational burden in huge datasets. Information Based Optimal Subdata Selection (IBOSS) [11] was proposed to find a subdata with maximal information matrix under D-optimality. The model is a deterministic method and was used in [12] by including logistic regression. A crucial problem faced by researchers dealing with deep learning models for forest fire is imbalanced datasets and the study of subsampling to balance datasets was useful in this research.

A study of Canadian forest fire statistics is important to predict forest fire weather. Groot et al. [13] concluded that the National Fire Data Base (NFDB) points and polygons give the best estimates of fire season start and end dates. Coogan et al. [14] found that natural processes like fires were

essential to maintain ecological integrity. Hence, coexistence with a forest fire and allowing prescribed burns were proposed. The limitation of previous studies is that they mostly focus on narrow geographic regions resulting in regional bias while making predictions on fire.

The model proposed in [15] was developed to predict the affected area of small and frequent fires using meteorological inputs: temperature, relative humidity, wind speed, and precipitation. Latifah et al. [16] aimed to evaluate Random Forest (RF) in predicting forest fires based on climate and satellite data. The present study proposes to overcome these limitations by spatio-temporal agnostic process.

The Canadian Fire Weather Index (FWI) has been extensively applied in this study, and other researches such as Stocks et al. [17] provided a good background to this research. Giovanni et al. [18] proposed a daily Fire Risk Index. Though the study region is Italy, the Canadian FWI was used and it showed that FWI was an effective system. A methodology for automatic calibration of the CFFWIS to measure fire danger for a specific region was proposed in [19]. In our study, the proposed models are trained to be effective in diverse environments without any need for re-calibration.

III. DATASETS

Two sources of data were used: Canadian National Fire Database (CNFDB) and Canadian Weather Energy and Engineering Datasets (CWEEDS).

The CNFDB dataset [20] provides historical information on fire events for the entire Canada and also includes information on the shape, time, cause, and location of the fire events. When making predictions using this dataset, only the features of importance as listed in Table I were used. An explanation of the data preprocessing is described in Section IV.

TABLE I
PREPROCESSED CNFDB FEATURES AND DESCRIPTION

Feature	Description
REP_DATE	Date associated with fire
OUT_DATE	Reported date of fire extinguished
CALC_HA	Fire size in hectares with higher precision
CAUSE	Specifies the cause of fire
GEOMETRY	Fire polygon

The CWEEDS dataset [21] is available for the period from 1998 to 2017 and has an hourly granularity. A document detailing the data can be found in [22]. This dataset provides a historical record of the climate/weather across continental Canada for multiple weather stations. It contains one file for all the weather station metadata and then each weather station data is contained in a separate file grouped by provinces in Canada. Table II lists the preprocessed features from the CWEEDS dataset.

IV. METHODOLOGY

This paper proposes to use similar features as in FWI to generate a generalized model that can be used anywhere in the world without calibration. In this attempt, there was a

TABLE II
PREPROCESSED CWEEDS FEATURES WITH UNITS

S.No.	Feature	Unit
1	Extraterrestrial irradiance	kJ/m^2
2	Global horizontal irradiance	kJ/m^2
3	Direct normal irradiance	kJ/m^2
4	Diffuse horizontal irradiance	kJ/m^2
5	Global horizontal illuminance	lux
6	Direct normal illuminance	lux
7	Diffuse horizontal illuminance	lux
8	Zenith luminance	Cd/m^2
9	Minutes of sunshine	minutes
10	Ceiling height	m
11	Sky layer 1	Digit codes
12	Sky layer 2	Digit codes
13	Sky layer 3	Digit codes
14	Sky layer 4	Digit codes
15	Visibility	km
16	Weather-Thunderstorm	Digit codes
17	Weather-Rain	Digit codes
18	Weather-Drizzle	Digit codes
19	Weather-Snow 1	Digit codes
20	Weather-Snow 2	Digit codes
21	Weather-Ice	Digit codes
22	Weather-Visibility 1	Digit codes
23	Weather-Visibility 2	Digit codes
24	Station pressure	Pa
25	Dry bulb temperature	$^{\circ}\text{C}$
26	Dew point temperature	$^{\circ}\text{C}$
27	Wind direction	degree
28	Wind speed	m/s
29	Total sky cover	Oktas
30	Opaque sky cover	Oktas
31	Snow cover	Boolean

difficulty in finding a balanced dataset. The datasets were found to have more than 100,000 non-fire points for 1 fire point leading to highly imbalanced datasets. There has been research to subsample datasets [9]–[12], but there has been none so far on forest fires. Initially, it was proposed to change the spatial and temporal resolution while using subsampling. However, such techniques will in turn necessitate re-calibration of the model when it is used in other regions. This led to solving two challenges: the first being to generate a generic deep learning model and, the second, a data subsampling framework that will not need calibration.

It was noted that the local weather conditions influence forest fires. Hence, all the models in this research were provided with time series input taking in the past N days for both fire and non-fire data points. Fires can occur anywhere with vegetation but the weather stations are spread unevenly across the country. Hence, another factor K that denotes the number of nearby weather stations was added to the testing. A K-Nearest Neighbour (KNN) clustering was used to get the weather stations. The value is averaged when $K > 1$ (see Algorithm 1).

To generate a balanced dataset, i.e., a dataset with a ratio of 1:1 fire and non-fire event, this paper proposes to use the past M months as the starting point of the time series data

for non-fire data points as shown in Algorithm 3. A detailed description of the process is depicted in Fig. 1 and described in Section IV-A.

In this process, the model learns the weather variations over the past M months that are likely to produce fire. This paper also proposes to have the data standardized. When the model learns from the standardized values, it is likely to have the same performance in other regions as well. This is because the model has learnt the variation in data that is causing the fire. To ensure that the proposed subsampling framework works to generate a generalized model, it is applied to two deep learning models, i.e., binary classification and regression models.

Fifty-four experiments were run for each model type leading to a total of 108 experiments. All the experiments used three hyper-parameters, N , K , and M with varying values. Three different values of N (7, 14, and 30) were used to see how the number of past days impacts the models' predictions. K has three different values as well (1, 3, and 5) to see if having more than one station as input helps in improving the model performance. Finally, for M , six different values (3, 5, 6, 7, 9, and 12) were used to see how the time difference between fire and non-fire points affects the model predictions. $M = 12$ is designed to be the most difficult test case. This checks how the models perform when there is little variation in input features.

A. Data Preprocessing

Figure 1 presents the flow chart for data preprocessing. Preprocessing started by collecting all the CWEEDS weather station data CSV files with the features in Units and expanding the grouped features, such as, sky condition. At the same time, temperature units are set to $^{\circ}\text{C}$, distance units are set to m (meters), wind speed units are set to m/s , and illumination units are set to lux . After this, the various CSV files are merged into one file. Then from the CWEEDS weather station metadata file only *ClimateID*, *Latitude*, and *Longitude* were retained. The weather station metadata is then joined with the merged weather station data. The join is made on the *ClimateID* and is saved into the CWEEDS database (S) for future queries. All the features listed in the CNFDB dataset, except the ones listed in Table I are removed and stored in the CNFDB database (F) for future queries.

Table III describes the variables used in Algorithms 1-4. The values of S , F , and K are provided as input to Algorithm 1. K nearest stations from the set of S are selected for every fire event (f) in the set F . When the value of $K > 1$, the feature values are averaged. This returns W as its output which contains data of the K nearest weather station for each fire point.

The value of N is selected and is provided as input to Algorithm 2 along with F and W from Algorithm 1. Algorithm 2 loops over all fire events in F . It first extracts the *REP_DATE* (the day fire started) mentioned in Table I and assigns it to T . The *FTS* data is then taken from $T - N$ up until T . This time series information is appended to E_f for all fires.

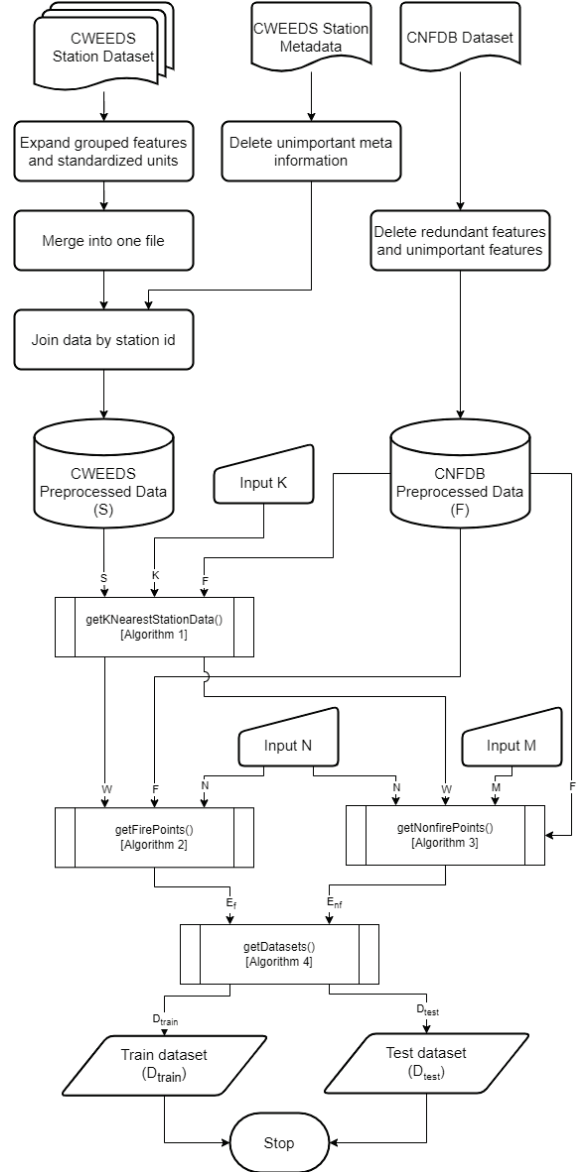


Fig. 1. Data Preprocessing Flow Chart

Algorithm 1 K-Nearest Weather Station Data Aggregation

All the symbols and notations used in the algorithm have been described in Table III.

Input: K, S, F

Output: W

```

1: procedure getKNearestStationData( $K, S, F$ )
2:    $W \leftarrow \text{Null}$ 
3:   for  $f \in F$  do
4:      $data_k \leftarrow \text{select } K \text{ nearest station data from } S$ 
5:      $data_{k\_avg} \leftarrow \text{average each feature value in } data_k$ 
6:      $W \leftarrow \text{push } data_{k\_avg} \text{ to } W$ 
7:   end for
8: end procedure

```

Now the value of M is selected and is provided as input to Algorithm 3 along with N , F and W from Algorithm 1.

TABLE III
ALGORITHMIC VARIABLE DESCRIPTION

Variable	Description
N	Number of past days
K	Number of nearest weather stations
M	Number of past months
S	Set of weather stations
F	Set of fire events
W	Weather data with spatio-temporal information
E_f	Subsampled weather dataset for fire events
E_{nf}	Subsampled weather dataset for non-fire events
D_{train}	Training dataset without spatio-temporal information
D_{test}	Testing dataset without spatio-temporal information
$dataset$	Intermediate dataset value
$dataset_c$	Dataset's column values
$data_k$	Subset containing K weather station data
$data_{k_avg}$	Subset containing average K weather station data
T	Date of fire event
T_{start}	Start date for time series data
T_{end}	End date for time series data
F_{TS}	Fire time series data
$N_{F_{TS}}$	Non-fire time series data
c	Column in dataset
μ	Mean
μ_c	Mean of column
σ	Standard deviation
σ_c	Standard deviation of column

Algorithm 2 Fire Point Extraction

All the symbols and notations used in the algorithm have been described in Table III.

Input: N, F, W
Output: E_f

```

1: procedure getFirePoints( $N, F, W$ )
2:    $E_f \leftarrow \text{Null}$ 
3:   for  $f \in F$  do
4:      $T \leftarrow f.REP\_DATE$ 
5:      $T_{start} \leftarrow T$ 
6:      $T_{end} \leftarrow (T - N)$ 
7:      $F_{TS} \leftarrow W$  data between  $[T_{start}, T_{end}]$ 
8:      $E_f \leftarrow \text{push } F_{TS} \text{ to } E_f$ 
9:   end for
10: end procedure
```

Algorithm 3 loops over all fire events in F . It first extracts the REP_DATE and assigns it to T . The $N_{F_{TS}}$ data is then considered from $T - M - N$ up until $T - M$. It is appended to E_{nf} and this process repeats for every fire event to obtain the required dataset.

The outputs from Algorithm 2 and Algorithm 3 are given as inputs to Algorithm 4. We append the appropriate target values to E_f and E_{nf} . In case of binary classification, E_f gets 1 and E_{nf} gets 0 while in case of regression, E_f gets the area burnt by fire event and E_{nf} gets 0. Both E_f and E_{nf} time series data are merged as one into $dataset$. From $dataset$, any spatial or temporal information is removed to make the data spatio-temporal agnostic. The $dataset$ is standardized using σ and μ as shown in steps 8 to 10 of Algorithm 4. Finally, the data is randomly sampled and split into D_{train} and D_{test} which are used for training and validation of deep learning models respectively. D_{train} gets 80% of the values from $dataset$ while D_{test} gets 20% of the values.

The final dataset will have slightly more fire data than non-

Algorithm 3 Non-Fire Point Extraction

All the symbols and notations used in the algorithm have been described in Table III.

Input: N, M, F, W
Output: E_{nf}

```

1: procedure getNonfirePoints( $N, M, F, W$ )
2:    $E_{nf} \leftarrow \text{Null}$ 
3:   for  $f \in F$  do
4:      $T \leftarrow f.REP\_DATE$ 
5:      $T_{start} \leftarrow (T - M)$ 
6:      $T_{end} \leftarrow (T - M - N)$ 
7:     if  $T_{start} \notin F$  then
8:        $N_{F_{TS}} \leftarrow W$  data between  $[T_{start}, T_{end}]$ 
9:        $E_{nf} \leftarrow \text{push } N_{F_{TS}} \text{ to } E_{nf}$ 
10:    end if
11:  end for
12: end procedure
```

Algorithm 4 Spatio-Temporal Agnostic Subsampling

All the symbols and notations used in the algorithm have been described in Table III.

Input: E_f, E_{nf}
Output: D_{train}, D_{test}

```

1: procedure getDataset( $E_f, E_{nf}$ )
2:    $dataset \leftarrow \text{Null}$ 
3:    $E_f \leftarrow \text{append target values to } E_f$ 
4:    $E_{nf} \leftarrow \text{append target values to } E_{nf}$ 
5:    $dataset \leftarrow \text{push } E_f \text{ to } dataset$ 
6:    $dataset \leftarrow \text{push } E_{nf} \text{ to } dataset$ 
7:    $dataset \leftarrow \text{delete spatio-temporal data from } dataset$ 
8:   for  $c$  in  $dataset$  do
9:      $dataset_c \leftarrow \frac{dataset_c - \mu_c}{\sigma_c}$ 
10:  end for
11:   $D_{train}, D_{test} \leftarrow \text{randomly partition the } dataset$ 
12: end procedure
```

fire data because sometimes there may have been forest fire M months ago also for the same location. Approximately, twenty thousand fire and non-fire time series records were obtained by applying this framework.

B. Binary Classification Modeling

The binary classification model had an activation of *ReLU* for all its hidden layers and input layer while having an activation of *Sigmoid* for the output layer. Binary Cross Entropy (BCE) was used as a loss function along with the Adam optimizer. In Algorithm 2 output (target value) was set to 1, while for Algorithm 3 the output was set to 0. The models are expected to learn the variation in weather features over the months M and act as a fire weather predictor that outputs the probability of fire occurrence. This paper hypothesizes that the spatio-temporal agnostic deep learning models will learn the features that are likely to cause a fire.

Figure 2a makes a visual representation of the deep learning binary classification model structure. The hidden layer size dropped 25% incrementally for each new hidden layer. Various learning rates were tested while developing the models. The learning rate of 0.00001 was found to have the best results. The models tend to converge soon, hence, a low number of epochs, that is 100, was chosen. *F1Score* is used to evaluate the models as it takes into account the precision and recall.

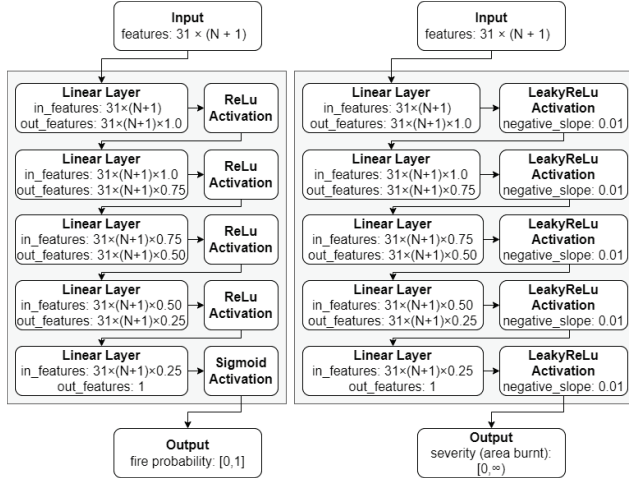


Fig. 2. a) Deep Learning Binary Classification Model Structure, b) Deep Learning Regression Model Structure

C. Regression Modeling

The regression model had an activation of *LeakyReLU* with a negative slope of 0.01 for all its layers. Mean Square Error (MSE) was used as a loss function along with the Stochastic Gradient Descent (SGD) optimizer. In Algorithm 2, output (target value) was set to the area burnt by the fire event while the output for Algorithm 3 was set to 0. The predicted values from the models can be used not only to predict the area but also to determine the severity of forest fires. The binary classification models will give the probability of fire, while the regression models will provide the severity of the fire. Combining the binary classification and regression models will give the fire weather behavior.

Figure 2b makes a visual representation of the deep learning regression model structure. In all the models, the layer sizes dropped 25% incrementally for each new hidden layer. Different learning rates and epochs were tested. It was found that a learning rate of 0.0001 was ideal for all the models. The models converge very slowly hence, a large number of epochs, that is 5000, was chosen. R^2 Score for regression models was used, since it tests for goodness of fit.

V. EXPERIMENTS AND RESULTS

Table II lists the 31 features which are obtained after being processed by Algorithm 4. These 31 features for $N + 1$ days were provided as the model input. There are $N + 1$ days since the present-day is included as well. Hence, each model had $31 \times (N + 1)$ input features. Since the initial parameters for all 54 experiments of the type binary classification and regression are the same, Table IV lists the comparison of the models' metrics with different combinations of N , K , and M for both binary classification and regression.

A. Binary Classification Results

It was considered a fire, if the model prediction was greater than 0.5. This gave us two events, one fire and the other non-fire. Column 5 ($F1Score$) in Table IV provides the $F1Score$

for each of the 54 binary classification model experiments. $F1Score$ is very high for 48 experiments. This shows that forest fire is highly dependent on weather features. In experiments, where M was set between 3 and 9, excellent $F1Score$ (0.9 to 1) was obtained. This accounted for 45 experiments. Three good $F1Score$ (0.8 to 0.9) for $M = 12$ (looking a year back in the same month) were obtained. Six acceptable $F1Score$ (0.70 to 0.8) for $M = 12$ were obtained. The test case $M = 12$ also performed well. This shows that the models can extrapolate fire weather patterns from weather data.

TABLE IV
REPRESENTATION OF MODEL METRICS WITH RESPECT TO K , N , AND M

Model Variation				Metrics	
N	K	M	no. of input features	$F1Score$	R^2Score
7	1	3	248	0.957	0.832
14	1	3	465	0.964	0.858
30	1	3	961	0.981	0.907
7	3	3	248	0.952	0.838
14	3	3	465	0.969	0.865
30	3	3	961	0.985	0.902
7	5	3	248	0.962	0.841
14	5	3	465	0.975	0.856
30	5	3	961	0.986	0.919
7	1	5	248	0.990	0.952
14	1	5	465	0.994	0.955
30	1	5	961	0.994	0.964
7	3	5	248	0.991	0.952
14	3	5	465	0.993	0.951
30	3	5	961	0.995	0.960
7	5	5	248	0.991	0.951
14	5	5	465	0.993	0.963
30	5	5	961	0.993	0.964
7	1	6	248	0.993	0.971
14	1	6	465	0.997	0.966
30	1	6	961	0.998	0.975
7	3	6	248	0.994	0.967
14	3	6	465	0.996	0.969
30	3	6	961	0.996	0.979
7	5	6	248	0.995	0.977
14	5	6	465	0.996	0.972
30	5	6	961	0.998	0.958
7	1	7	248	0.994	0.958
14	1	7	465	0.994	0.967
30	1	7	961	0.994	0.964
7	3	7	248	0.994	0.961
14	3	7	465	0.997	0.965
30	3	7	961	0.995	0.967
7	5	7	248	0.992	0.967
14	5	7	465	0.995	0.965
30	5	7	961	0.997	0.973
7	1	9	248	0.976	0.876
14	1	9	465	0.976	0.889
30	1	9	961	0.988	0.936
7	3	9	248	0.975	0.883
14	3	9	465	0.979	0.900
30	3	9	961	0.989	0.931
7	5	9	248	0.970	0.899
14	5	9	465	0.983	0.914
30	5	9	961	0.991	0.931
7	1	12	248	0.716	0.159
14	1	12	465	0.754	0.186
30	1	12	961	0.787	0.264
7	3	12	248	0.723	0.188
14	3	12	465	0.789	0.253
30	3	12	961	0.803	0.347
7	5	12	248	0.760	0.191
14	5	12	465	0.801	0.255
30	5	12	961	0.813	0.342

a) Binary Classification Model Metrics in Column $F1Score$, b) Regression Model Metrics in Column R^2Score

Figure 3 shows the impact of N (blue), K (orange), and M (yellow) on $F1Score$. The y-axis in these figures represents

the $F1Score$. The scale is set between 0.7 to 1. We see that N impacts the $F1Score$. Taking larger time series data as input will help with improved model performance. K has a minimal impact, this implies that we can take values from the nearest station only. M has a major impact on the $F1Score$ and performed best between values 5 to 7.

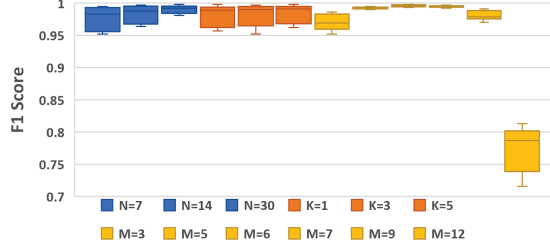


Fig. 3. Binary Classification Models: $F1Score$ vs N, K , and M

B. Regression Results

Column 6 (R^2Score) in Table IV provides the R^2Score for each of the 54 regression model experiments predicting the area burnt by the fire. For values, other than $M = 12$, the R^2Score is high and has a significant amount of variance explained. For values $M = 12$, the models did not perform optimally. Four of the nine models for $M = 12$ have little to no variance while five of them have some variance in the input features. The Binary Classification models were able to classify fire and non-fire events as they were bounded between 0 and 1. The regression models are not bounded. Rather they have a minimum value of zero but no upper bounds. Hence, the models had difficulty predicting the severity (area burnt).

Figure 4 shows the impact of K , N , and M respectively, on R^2Score . It can be seen that K has minimal impact, implying that we can take values from the nearest station only. N impacts the R^2Score , implying taking larger time series data as input will help with improved model performance. M has a major impact and performed best between values 5 to 7. We can predict the area burnt as long as there is a significant variance in the input features.

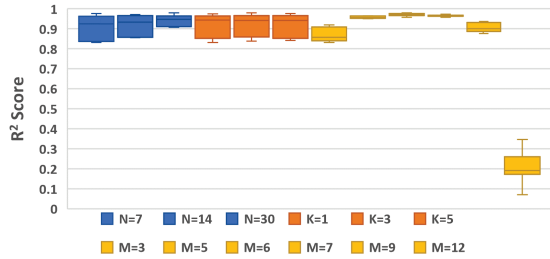


Fig. 4. Regression Models: R^2Score vs N, K , and M

VI. CONCLUSION

This paper provides a viable framework for data subsampling, as presented in Algorithm 4. This framework was successfully trained and tested on binary classification and regression models. It even works with limited variation between fire and non-fire data. Using this framework, 45 of the

54 binary classification model experiments built produced an $F1Score$ greater than 0.95 while 35 of 54 regression model experiments produced an R^2Score greater than 0.91.

Possible threats to validity include selecting different model hyper-parameters and improper data preprocessing. Acquisition of data in a framework different from the one specified in the paper will also potentially affect validating the results presented. As part of our future work, it is proposed to add vegetation data to improve its performance. Further validation of the framework, taking into consideration climate change, will also be helpful. We also plan to train the models with more values for M and N in our future work.

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